

LECTURE 3

Dynamic Regression, Error Correction and Exogeneity

Prof. Pedro Valls Pereira pedro.valls@fgv.br

FGV EAESP · MSc / PhD in Business Administration

Agenda

- 01 Autoregressive distributed lag (ADL) models
- 02 From ADL to the Error Correction Model
- 03 Engle–Granger single-equation cointegration test
- 04 Long-run and short-run dynamics
- 05 Weak exogeneity
- 06 Strong exogeneity
- 07 Super-exogeneity and policy invariance
- 08 Business applications and empirical practice

UK demand for money (velocity and interest rates, 1963–1990) from Hendry & Ericsson (1991) illustrates ECM, exogeneity, and policy analysis throughout the lecture.

Section 1

Autoregressive Distributed Lag Models

The ADL(p, q) Model I

$$Y_t = \alpha_0 + \sum_{i=1}^p \alpha_i Y_{t-i} + \sum_{j=0}^q \beta_j X_{t-j} + \varepsilon_t$$

Y_t : dependent variable; X_t : conditioning variable (weakly exogenous); $\varepsilon_t \sim \text{iid}(0, \sigma^2)$.

Why start from ADL?

- Nests static regression (set $p = q = 0$, β_0 only) and pure AR
- Residuals are white noise by construction if p, q chosen well
- Valid inference under general serial correlation in Y_t
- Basis for **general-to-specific** modelling (Lecture 7)
- Provides natural reparameterisation as an ECM

The ADL(p, q) Model II

$$Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \beta_0 X_t + \beta_1 X_{t-1} + \varepsilon_t$$

This single equation already captures:

- Short-run effect of X : β_0
- Long-run effect of X : $(\beta_0 + \beta_1)/(1 - \alpha_1)$
- Speed of adjustment: $1 - \alpha_1$

The ADL(p, q) Model III

Long-run solution of ADL(1,1)

In the long run $Y_t = Y_{t-1} = Y^*$ and $X_t = X_{t-1} = X^*$:

$$Y^* = \frac{\alpha_0}{1 - \alpha_1} + \frac{\beta_0 + \beta_1}{1 - \alpha_1} X^* = \mu + \theta X^*$$

where $\theta = (\beta_0 + \beta_1)/(1 - \alpha_1)$ is the **long-run multiplier**.

Impact multiplier

Effect of a unit increase in X_t on Y_t in the **same period**: β_0

Interim multiplier (lag s)

Cumulative effect after s periods. Converges to θ as $s \rightarrow \infty$ if $|\alpha_1| < 1$.

The ADL(p, q) Model IV

The long-run relationship is well-defined **only if** $|\alpha_1| < 1$ (stationarity condition). If $\alpha_1 = 1$ (unit root in Y_t), we need cointegration to interpret θ .

Section 2

From ADL to the Error Correction Model

The ECM Reparameterisation I

Subtract Y_{t-1} from both sides of ADL(1,1):

$$\Delta Y_t = \beta_0 \Delta X_t - (1 - \alpha_1) \underbrace{[Y_{t-1} - \theta X_{t-1} - \mu]}_{\text{error correction term}} + \varepsilon_t$$

- $\beta_0 \Delta X_t$: short-run response to changes in X
- $(1 - \alpha_1)$: **speed of adjustment** back to long-run equilibrium
- $[Y_{t-1} - \theta X_{t-1} - \mu]$: **disequilibrium error** — how far the system was from equilibrium last period
- If $Y_{t-1} > \theta X_{t-1} + \mu$ (above equilibrium), ΔY_t is pulled **downward**

If Y_t and X_t are both $I(1)$ and cointegrated, they **always** have an ECM representation, and vice versa. Cointegration $\Leftrightarrow$ ECM.

General ECM from $ADL(p, q)$

$$\Delta Y_t = \sum_{i=1}^{p-1} \tilde{\alpha}_i \Delta Y_{t-i} + \beta_0 \Delta X_t + \sum_{j=1}^{q-1} \tilde{\beta}_j \Delta X_{t-j} - \underbrace{\pi [Y_{t-1} - \theta X_{t-1}]}_{ECT} + \varepsilon_t$$

where $\pi = -(1 - \alpha_1 - \dots - \alpha_p)$ is the adjustment coefficient (should be negative for stability) and θ is the long-run multiplier.

Identifying the long-run — three equivalent approaches:

- 1 Estimate ADL, compute $\theta = \sum \hat{\beta}_j / (1 - \sum \hat{\alpha}_i)$
- 2 Run static OLS of Y_t on X_t (Engle–Granger step 1)
- 3 Impose $\pi\theta$ directly and estimate jointly

The ECM Reparameterisation III

Differencing all variables and estimating $\Delta Y_t = \beta_0 \Delta X_t + \varepsilon_t$ throws away the long-run information. The ECM preserves both short-run dynamics **and** long-run equilibrium.

Section 3

Engle–Granger Cointegration Test

Single-Equation Cointegration Testing I

Engle–Granger two-step procedure (1987):

Step 1: Estimate the long-run (cointegrating) regression by OLS:

$$Y_t = \mu + \theta X_t + \hat{u}_t$$

Step 2: Test whether $\hat{u}_t \sim I(0)$ using an ADF-type test on the residuals:

$$\Delta \hat{u}_t = \pi \hat{u}_{t-1} + \sum_{j=1}^k \gamma_j \Delta \hat{u}_{t-j} + \eta_t$$

Test $H_0 : \pi = 0$ (non-cointegration) vs. $H_1 : \pi < 0$ (cointegration).

Because $\hat{u}_t$ is an estimated residual (not observable), the critical values are **more negative** than standard DF values. They depend on the number of variables and whether a constant/trend is included. Use MacKinnon (1991) response surface critical values.

Advantages of EG two-step

- Simple OLS estimation
- Yields the long-run coefficient directly
- Easy to compute adjustment speed from ECM in step 2
- Numerically robust

Limitations

- Assumes **one** cointegrating vector (valid with 2 variables, restrictive with more)
- Super-consistent ($\sqrt{T}$ rate) but biased in finite samples
- Sensitive to which variable is normalised on the left-hand side
- For multiple variables $\Rightarrow$ use Johansen (Lecture 4)

Single-Equation Cointegration Testing III

OLS in a cointegrating regression is **super-consistent**: $\hat{\theta} \rightarrow \theta$ at rate T rather than $\sqrt{T}$. This means that even though Y_t and X_t are $I(1)$, the OLS estimator of θ is consistent and converges faster than usual. However, it is **not** asymptotically normal — use DOLS or FMOLS for inference on θ .

Section 4

Weak, Strong, and Super-Exogeneity

The Engle–Hendry–Richard (1983) Framework I

Setup: Data $X_t = (Y_t, Z_t)'$. We wish to model Y_t conditional on Z_t .

Z_t is **weakly exogenous** for ψ (parameters of interest) if:

- 1 The joint density factors as $f(Y_t, Z_t) = f(Y_t|Z_t) \cdot f(Z_t)$
- 2 The parameters of interest ψ appear only in $f(Y_t|Z_t)$, not in $f(Z_t)$
- 3 ψ and λ (parameters of marginal model for Z_t) are **variation-free**

Implication: Conditioning on Z_t involves no loss of information about ψ . Valid to estimate the conditional model without modelling Z_t .

Weak exogeneity concerns the **parameters** of the model. Granger non-causality concerns the **distribution** (whether past Y helps predict Z). They are different concepts and each can hold without the other.

The Engle–Hendry–Richard (1983) Framework II

Z_t is **strongly exogenous** for ψ if it is **weakly exogenous** for ψ **and** Y_t does not Granger-cause Z_t .

Implication: Valid for both estimation (uses only conditional model) **and** multi-step forecasting (can construct future paths of Z_t independently of Y_t).

Z_t is **super-exogenous** for ψ if it is **weakly exogenous** for ψ **and** the parameters ψ in the conditional model are **invariant** to changes (interventions) in the marginal model of Z_t .

Implication: The conditional model remains valid even when the policy regime governing Z_t changes — the structural parameters ψ are **policy-invariant**. Crucial for the **Lucas critique**.

The Engle–Hendry–Richard (1983) Framework III

The Lucas Critique and Super-Exogeneity

Lucas (1976): if agents' expectations adapt to policy changes, then econometric models estimated under one policy regime will fail when the regime changes — parameters are not structural.

Super-exogeneity is the answer:

- If Z_t is super-exogenous for ψ in the $Y|Z$ model, then ψ is **invariant** to changes in the process generating Z_t
- Policy interventions that affect the marginal model of Z_t do **not** affect ψ
- This validates the conditional model for policy analysis

- 1 Detect breaks in the marginal model of Z_t using dummies D_t
- 2 Add those same D_t to the conditional model for $Y_t|Z_t$
- 3 If dummies are **insignificant** in the conditional model $\Rightarrow$ parameters are invariant $\Rightarrow$ support for super-exogeneity

The Engle–Hendry–Richard (1983) Framework IV

UK money demand example (Hendry & Ericsson, 1991)

Model: $\Delta v_t = f(\Delta R_{la,t}, ecm_{t-1}, \text{dummies})$

where $v_t = \log(M1/P)_t$ is velocity and $R_{la,t}$ is the local authority rate (cost of holding money).

- **Structural break** in $R_{la,t}$ after 1985 (own rate of money changed from zero)
- **Weak exogeneity** of $R_{la,t}$: adjustment coefficients appear only in velocity equation, not in interest rate equation
- **Super-exogeneity** test: add interest rate dummies to velocity equation $\Rightarrow$ insignificant $\Rightarrow$ velocity equation is structurally stable
- **Implication**: The model can be used for policy analysis even though the interest rate regime changed

When Y_t and Z_t both experience breaks, but a linear combination has no break, this is **co-breaking**. It is the analogue of cointegration for structural breaks and is related to super-exogeneity.

Section 5

Business Applications

Long-run advertising-sales relationship

If both sales S_t and advertising spend A_t are $I(1)$:

- 1 Test for cointegration (EG step 1)
- 2 If cointegrated, estimate ECM:

$$\Delta S_t = \beta_0 \Delta A_t - \pi \underbrace{[S_{t-1} - \theta A_{t-1}]}_{\text{ECT}} + \varepsilon_t$$

- 3 θ : long-run advertising elasticity
- 4 π : how fast sales adjust to new equilibrium
- 5 β_0 : immediate (quarter-on-quarter) response

Demand systems with prices

ECM for quantity demanded given price:

$$\Delta q_t = \beta_0 \Delta p_t - \underbrace{\pi [q_{t-1} - \theta p_{t-1}]}_{\text{ECT}} + \varepsilon_t$$

θ : long-run price elasticity (typically negative). π : adjustment speed (how quickly market clears).
Price exogeneity: test whether p_t responds to demand disequilibrium. If not $\Rightarrow$ weakly exogenous.

Applied checklist for ECM modelling

1. **Test each series for unit roots.** Use ADF + KPSS.
2. **Test for cointegration.** Engle–Granger residual-based test. Critical values from MacKinnon (1991).
3. **Estimate the ADL and reparameterise as ECM.** Verify that the adjustment coefficient $\hat{\alpha} < 0$ and significant.
4. **Test weak exogeneity.** Check that the conditioning variable Z_t does not load on the error correction term.
5. **Test super-exogeneity.** Add break dummies (detected in marginal model of Z_t) to the conditional model.
6. **Diagnostic checking.** Residuals: white noise (Ljung–Box), normality, ARCH, recursive parameter stability (CUSUM).

Core references

- Engle, R.F. & Granger, C.W.J. (1987). Co-integration and error correction. *Econometrica*, 55, 251–276.
- Engle, R.F., Hendry, D.F. & Richard, J.-F. (1983). Exogeneity. *Econometrica*, 51, 277–304.
- Hendry, D.F. & Doornik, J.A. (2014). *Empirical Model Discovery*. MIT Press.
- Pesaran, M.H. (2015). *Time Series and Panel Data Econometrics*. OUP. Chap. 21.

Applied papers

- Hendry, D.F. & Ericsson, N.R. (1991). Modeling narrow money demand in the UK and US. *European Economic Review*, 35, 833–881.
- Dubé, J.-P., Hitsch, G.J. & Manchanda, P. (2005). An empirical model of advertising dynamics. *QME*, 3, 107–144.
- Dekimpe, M.G. & Hanssens, D.M. (1995). The persistence of marketing effects. *Marketing Science*, 14, 1–21.

Software: R (`dynlm`, `urca`, `ca.po`, `gets`) or Python (`statsmodels.tsa.ardl`). Applied exercises: estimate an ADL for a marketing or financial series, test cointegration, reparameterise as ECM, test exogeneity.

Next: Lecture 4

Multivariate Dynamics: VAR, Cointegration and VEC

VAR(p) · Granger causality · Impulse responses · Johansen tests · VECM